447626

Course Name: Database Management Systems

Course Outcome

CO1-Understand the concept of database management systems and Relational database.
CO2- Identify the various data model used in database design.
CO3- Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4- Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5- Select the information from a database by formulating complex queries in SQL.
CO6- Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7- Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8- Discuss recovery system and be familiar with introduction to web database, distributed databases.

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Mid Term Examination, Odd Semester 2024-25
B.TECH. (CS), Year: II, Semester: III
BCSC 1003, Database Management Systems

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all sections
2. Use only blue pen to answer.
3. No B sheet will be given.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Amazon online is planning to develop a new database system to efficiently manage its academic records, including student information, course schedules, and faculty data. As part of the team responsible for this project, you are tasked with gathering requirements to ensure the system meets the needs of all stakeholders. What steps would you take in the Database Development Life Cycle (DDLC) to accurately gather these requirements?	3	1	A	F
2	Briefly explain the following with suitable example: i. Domain Constraints ii. Referential Integrity Constraints iii. Entity Integrity Constraint	3	4	U	F
3	Let Relation R = (A, B, C, D, E) be a relation scheme with the following Functional dependencies- AB→C C → D B → E Determine all possible candidate keys and super keys.	3	2	An	P

4	Consider a database relation named EMPLOYEE_PROJECT that contains the attributes EmployeeID, ProjectID, Role, and Salary. The following functional dependencies are observed in this relation: {EmployeeID, ProjectID → Role} {EmployeeID, ProjectID → Salary} {EmployeeID → Role} {Role → Salary} Analyze the functional dependencies and discuss the highest normal form.	3	3	U	F
5	Imagine you are tasked with designing a database for a large public library system. The library has thousands of books, including multiple copies of the same title, each of which must be uniquely identifiable. Patrons should be able to search for books by various attributes such as title, author, and genre, while the system should also track the availability and condition of individual copies. What database model would you choose to best manage this system, and how would you design the schema to ensure each book and its copies are uniquely identified and easily searchable?	3	6	E	C

Section – B

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Suppose a relational schema R = (P, Q, R, S), and set of functional dependency as following: F : { P→ QR, Q →R, P →Q, PQ →R } Find the canonical cover F_c (Minimal set of functional dependency).	5	6	E	C
7	Draw and explain 3-tier Architecture to each tier and write short notes on: 1. Specialization 2. Generalization	5	5	C	C
8	Consider the following relation schema Works (Pname,Cname,salary) Lives (Pname,Street,City)	5	6	An	P

	Located_in (Cname, city) Manager (Pname, Mgrname) Write the relational algebra queries for the following: a). Find the names of all persons who live in the city Bangalore. b). Retrieve the names of all person of "Infosys" whose salary is between Rs .50000 c). Find the names of all persons who lives and work in the same city d). List the names of the people who work for "Tech M" along with the cities they live in. e). Find the average salary of "Infosys" persons				